Quantification: 877_wt_z087c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 12 |
| interior lacuna count | 8 |
| frame-edge lacuna count | 4 |

Interior lacunae only: mean, median and sample SD (n = 8)
---------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 4121.67 | 4349 | 1874.68 | px^2 |
| major_axis_length_px | lacuna shape | 109.759 | 118.557 | 22.0224 | px |
| minor_axis_length_px | lacuna shape | 50.0334 | 51.7906 | 15.8766 | px |
| aspect_ratio | lacuna shape | 2.3302 | 2.1036 | 0.5642 | unitless |
| eccentricity | lacuna shape | 0.891 | 0.8795 | 0.0377 | unitless |
| solidity | lacuna shape | 0.7983 | 0.8195 | 0.0779 | unitless |
| roots_count | headline | 11.5 | 13 | 7.329 | count |
| ring_length_r30_px | headline | 383.5 | 400.5 | 220.995 | px |
| ring_length_r60_px | reported | 940.875 | 1105 | 500.73 | px |
| owned_length_px | ownership-dependent | 1955.13 | 2128.34 | 1279.16 | px |
| edge_count | ownership-dependent | 93.875 | 111 | 62.1138 | count |
| mean_edge_length_px | ownership-dependent | 20.7274 | 20.5054 | 2.07 | px |
| perimeter_px | normalised | 326.178 | 335.869 | 106.391 | px |
| ring_area_r30_px2 | normalised | 10933.1 | 11032 | 2659.36 | px^2 |
| ring_area_r60_px2 | normalised | 26665 | 27035 | 5699.49 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9993 | 1 | 0.002 | unitless |
| in_frame_fraction_r60 | normalised | 0.9826 | 1 | 0.0493 | unitless |
| ring_density_r30 | normalised | 0.0330377 | 0.0378335 | 0.0149541 | px^-1 |
| ring_density_r60 | normalised | 0.0334097 | 0.0390407 | 0.0159625 | px^-1 |
| roots_per_100px_perimeter | normalised | 3.3601 | 3.798 | 1.9841 | per 100 px |
| ring_attached_length_r30_px | network v2 | 342.5 | 366 | 225.096 | px |
| ring_attached_length_r60_px | network v2 | 767 | 879 | 481.389 | px |
| ring_length_w_r30_px | network v2 | 441.562 | 455.611 | 255.009 | px |
| ring_length_w_r60_px | network v2 | 1081.41 | 1269.69 | 580.843 | px |
| sholl_crossings_r10 | network v2 | 9.25 | 10 | 5.2847 | count |
| sholl_crossings_r20 | network v2 | 12.125 | 11 | 6.8544 | count |
| sholl_crossings_r30 | network v2 | 13.75 | 15 | 6.6494 | count |
| loop_count | loops, report only | 8.5 | 11 | 6.8661 | count |
| loop_edge_share | loops, report only | 0.2327 | 0.3051 | 0.177 | share |
| ring_width_mean_r30_px | width | 6.9027 | 6.9112 | 0.8585 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.009e+06 | px^2 |
| total_skeleton_length_px | 30977 | px |
| canalicular_length_density_per_px | 0.0307008 | px^-1 |
| junction_count | 630 | count |
| junction_density_per_px2 | 0.00062438 | px^-2 |
| lacuna_count | 12 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.189e-05 | px^-2 |
| skeleton_component_count | 391 | count |
| mean_component_length_px | 79.2251 | px |
| median_component_length_px | 23 | px |
| field_length_density_w_per_px | 0.0350975 | px^-1 |
| field_density_without_flagged_per_px | 0.0307008 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 716 | px |
| unattached_component_count | 36 | count |
| canalicular_length_density_attached_per_px | 0.0307008 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0314105 | px^-1 |
| excluded_area_px2 | 0 | px^2 |
| analysed_field_area_px2 | 1.04858e+06 | px^2 |
| loop_count | 87 | count |
| loop_edge_share | 0.1932 | share |
| edge_count | 1615 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.3231 | px |
| width_median_px | 8 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 30977 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1757 | 77.0533 | 35.9762 | 2.1418 | 0.8843 | 0.7142 | 1.417 | 13.6949 | 531.608 | True |
| 2 | 4317 | 116.364 | 50.6241 | 2.2986 | 0.9004 | 0.8545 | 0.8274 | 37.7415 | 814.323 | True |
| 3 | 1047 | 73.2742 | 21.411 | 3.4223 | 0.9564 | 0.675 | 0.4727 | 102.495 | 237.658 | False |
| 4 | 5980 | 133.077 | 65.0796 | 2.0448 | 0.8723 | 0.7429 | 0.7964 | 208.483 | 500.988 | False |
| 5 | 4142 | 119.005 | 48.2749 | 2.4652 | 0.914 | 0.8826 | 0.2092 | 399.882 | 20.3518 | True |
| 6 | 4805 | 118.959 | 55.0153 | 2.1623 | 0.8866 | 0.8357 | 0.4696 | 413.392 | 307.567 | False |
| 7 | 3114 | 93.6668 | 46.7845 | 2.0021 | 0.8663 | 0.8167 | 0.8128 | 455.325 | 614.1 | False |
| 8 | 5891 | 121.42 | 63.3439 | 1.9168 | 0.8531 | 0.8972 | 1.4767 | 461.976 | 812.6 | False |
| 9 | 1692 | 71.502 | 34.0722 | 2.0985 | 0.8792 | 0.9082 | 0.0325 | 636.572 | 12.6684 | True |
| 10 | 3893 | 118.155 | 48.5658 | 2.4329 | 0.9116 | 0.8222 | 0.9667 | 739.935 | 833.772 | False |
| 11 | | | | | | | | 860.708 | 550.499 | False |
| 12 | | | | | | | | 970.308 | 685.063 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 9 | 343 | 732 | 1267.79 | 70 | 18.1112 | 7.5173 |
| 2 | 8 | 245 | 592 | 942.034 | 39 | 24.1547 | 7.215 |
| 3 | 13 | 374 | 1029 | 1888.16 | 97 | 19.4656 | 8.1759 |
| 4 | 23 | 761 | 1623 | 3410.19 | 164 | 20.7938 | 7.6347 |
| 5 | 5 | 215 | 584 | 438.262 | 28 | 15.6522 | 6.7369 |

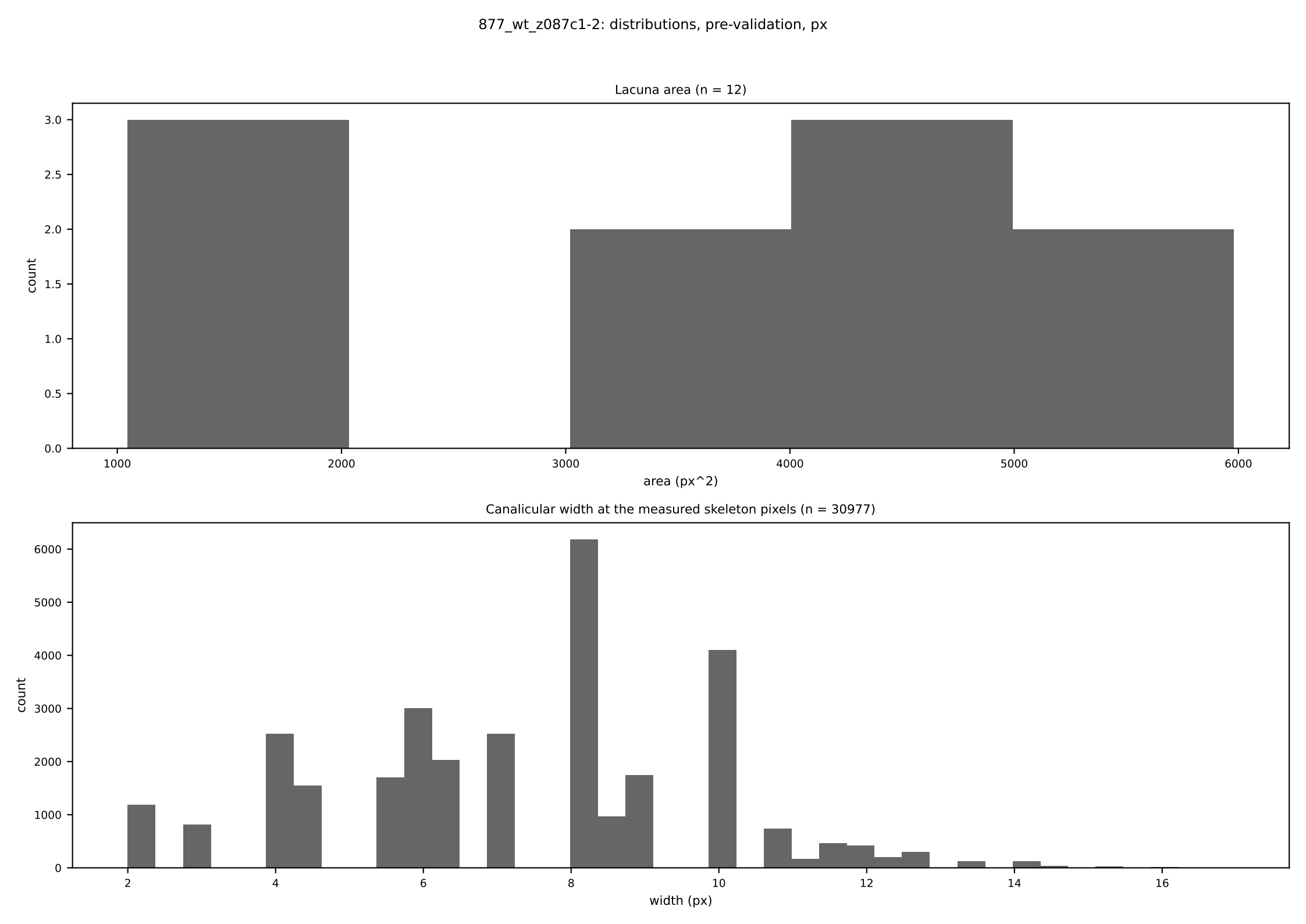
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 13 | 468 | 1125 | 2341.59 | 125 | 18.7327 | 7.2639 |
| 7 | 12 | 394 | 1120 | 2903.15 | 142 | 20.4447 | 6.7385 |
| 8 | 14 | 494 | 1090 | 3040.56 | 140 | 21.7183 | 7.0839 |
| 9 | 6 | 136 | 381 | 316.865 | 13 | 24.3742 | 7.9135 |
| 10 | 15 | 407 | 1154 | 1915.08 | 76 | 25.1985 | 6.5761 |
| 11 | 1 | 108 | 315 | 123.397 | 6 | 20.5662 | 6.4177 |
| 12 | 1 | 62 | 71 | 18.8995 | 1 | 18.8995 | 5.3312 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 274.066 | 6033 | 14351 | 0.6294 | 0.5895 | 0.056854 | 0.0510069 | 3.2839 |
| 2 | 340.475 | 9033 | 20414 | 0.7674 | 0.7052 | 0.0271228 | 0.0289997 | 2.3497 |
| 3 | 206.551 | 8300 | 22062 | 1 | 1 | 0.0450602 | 0.0466413 | 6.2938 |
| 4 | 502.186 | 15464 | 35653 | 1 | 1 | 0.0492111 | 0.0455221 | 4.58 |
| 5 | 336.628 | 7317 | 17241 | 0.6107 | 0.5856 | 0.0293836 | 0.0338728 | 1.4853 |
| 6 | 357.404 | 12357 | 30057 | 1 | 1 | 0.0378733 | 0.0374289 | 3.6373 |
| 7 | 314.333 | 10425 | 25683 | 1 | 1 | 0.0377938 | 0.0436086 | 3.8176 |
| 8 | 364.718 | 12512 | 29886 | 1 | 1 | 0.0394821 | 0.0364719 | 3.8386 |
| 9 | 190.953 | 4836 | 12429 | 0.5823 | 0.5596 | 0.0281224 | 0.0306541 | 3.1421 |
| 10 | 397.002 | 11639 | 28387 | 1 | 1 | 0.0349686 | 0.0406524 | 3.7783 |
| 11 | 301.404 | 9757 | 24715 | 1 | 1 | 0.011069 | 0.0127453 | 0.3318 |
| 12 | 165.823 | 7011 | 16877 | 0.9943 | 0.8606 | 0.00884325 | 0.00420691 | 0.6031 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 311 | 618 | 395.363 | 846.252 | 8 | 7 | 10 |
| 2 | 227 | 455 | 289.492 | 684.896 | 8 | 9 | 11 |
| 3 | 349 | 965 | 422.777 | 1160.41 | 9 | 10 | 14 |
| 4 | 734 | 1490 | 878.394 | 1882.16 | 16 | 21 | 22 |
| 5 | 153 | 321 | 240.823 | 658.742 | 6 | 6 | 7 |
| 6 | 413 | 930 | 554.441 | 1309.78 | 13 | 18 | 19 |
| 7 | 332 | 874 | 448.99 | 1289.76 | 10 | 11 | 16 |
| 8 | 448 | 869 | 565.943 | 1249.63 | 13 | 19 | 18 |
| 9 | 134 | 281 | 154.711 | 424.848 | 5 | 4 | 7 |
| 10 | 383 | 884 | 462.233 | 1324.11 | 10 | 11 | 13 |
| 11 | 64 | 107 | 130.024 | 358.333 | 2 | 6 | 4 |
| 12 | 17 | 17 | 69.6985 | 77.1127 | 1 | 1 | 4 |


877_wt_z087c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 30977)
count
width (px)